

SEM NAN, Iran, Islamic Republic of

WMO: 407570

Lat: 35.5878N

Lon: 53.4216E

Elev: 3698

StdP: 12.84

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	28.1	6.5	8.8	48.7	10.1	10.6	46.0	18.4	40.3	14.2	42.7	1.5	330	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	104.0	64.5	101.8	64.2	99.6	63.8	69.2	96.1	67.5	95.0	66.2	93.5	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.3	86.4	85.5	57.0	79.5	84.3	54.9	73.6	83.3	35.7	95.9	34.2	95.1	33.0	94.0	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.7	17.4	14.2	21.6	107.3	4.5	1.3	18.3	108.3	15.7	109.0	13.2	109.8	9.9	110.7		
			19.5	71.2	3.8	3.7	16.8	73.9	14.6	76.0	12.5	78.1	9.7	80.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.7	39.8	44.7	54.6	64.6	76.2	85.9	90.3	87.2	80.2	67.8	52.5	42.7
	DBStd	18.65	5.95	6.70	7.74	7.37	5.46	5.04	4.59	4.20	4.76	6.27	7.07	5.22
	HDD50	819	319	168	40	4	0	0	0	0	0	0	54	234
	HDD65	2889	782	569	332	92	1	0	0	0	0	44	376	693
	CDD50	6531	2	19	184	441	812	1078	1248	1154	907	551	129	6
	CDD65	3127	0	0	11	79	348	628	783	689	457	130	2	0
	CDH74	42196	0	1	88	668	3663	9103	12199	10099	5312	1055	8	0
	CDH80	24048	0	0	17	171	1533	5412	7939	6077	2613	286	0	0
Wind	WSAvg	4.8	3.0	3.9	4.7	5.6	6.3	6.9	6.7	5.9	4.9	4.0	3.1	2.7
Precipitation	PrecAvg	8.1	0.9	1.0	1.3	1.2	0.7	0.3	0.4	0.2	0.3	0.3	0.7	0.8
	PrecMax	12.6	2.8	2.6	3.6	3.1	2.5	1.9	1.7	1.2	1.5	1.3	1.9	1.8
	PrecMin	4.2	0.3	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	2.0	0.6	0.7	0.9	0.7	0.6	0.4	0.4	0.3	0.3	0.3	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	68.2	82.5	89.8	95.4	104.3	107.3	105.3	98.7	89.6	75.0	62.4
		MCWB	44.7	47.8	53.2	57.0	59.7	64.0	64.9	64.2	61.0	59.3	53.4	47.4
	2%	DB	55.7	62.3	75.4	84.6	93.2	101.5	104.9	102.3	96.1	86.0	70.7	57.5
		MCWB	43.2	44.9	51.3	55.8	59.1	63.3	64.7	64.6	61.2	57.7	52.7	44.6
	5%	DB	52.8	59.1	71.5	80.7	90.8	99.3	102.7	100.1	93.6	83.2	67.6	54.9
		MCWB	41.6	43.8	49.6	54.3	58.5	62.3	64.5	64.3	60.7	56.7	51.0	43.8
	10%	DB	49.8	55.9	67.9	77.3	88.0	97.2	100.6	97.9	91.5	80.3	64.0	51.9
		MCWB	40.2	42.8	48.1	53.3	57.9	61.8	64.5	63.7	60.4	55.5	48.9	42.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	50.1	55.0	59.3	64.5	70.4	71.9	70.9	67.5	62.9	59.7	48.9
		MCDB	55.3	58.7	71.4	79.0	89.8	95.0	95.6	97.3	89.6	82.9	67.8	57.9
	2%	WB	44.7	47.3	52.9	57.6	61.3	66.5	69.7	68.5	65.2	59.4	54.8	46.8
		MCDB	52.3	59.3	69.4	77.8	87.4	94.2	97.3	95.0	88.3	80.8	65.7	55.2
	5%	WB	42.6	45.4	51.4	56.3	60.1	64.9	68.0	67.1	63.3	57.8	52.2	45.1
		MCDB	50.8	56.5	68.1	75.7	85.7	92.5	96.3	93.7	87.9	79.2	63.7	52.6
	10%	WB	40.9	43.7	49.7	54.9	59.0	63.6	66.6	65.8	62.0	56.6	50.4	43.2
		MCDB	48.8	54.3	65.0	73.3	83.2	91.8	95.1	92.2	86.9	77.0	61.8	50.0
Mean Daily Temperature Range	5% DB	MDBR	15.5	17.1	18.6	19.2	19.6	20.1	18.7	20.2	20.6	19.2	16.4	15.4
		MCDBR	18.4	20.1	22.1	21.2	21.2	20.8	19.8	21.1	22.0	21.5	19.0	18.1
		MCWBR	10.9	10.4	10.1	8.5	7.9	7.4	6.5	7.7	9.0	9.3	10.4	10.8
	5% WB	MCDBR	17.0	18.2	19.2	19.4	20.0	20.2	18.9	19.0	20.5	20.1	17.1	16.2
		MCWBR	10.7	10.3	9.7	8.4	7.9	8.3	7.6	7.6	9.2	9.5	10.4	10.3
Clear-Sky Solar Irradiance	taub		0.340	0.384	0.459	0.539	0.552	0.536	0.572	0.520	0.465	0.447	0.385	0.337
	taud		2.218	2.076	1.880	1.741	1.739	1.799	1.740	1.853	1.958	1.976	2.110	2.246
	Ebn at Noon		266	266	254	241	240	243	233	243	250	240	245	259
	Edh at Noon		34	44	59	72	73	69	72	63	54	49	38	31
All-Sky Solar Radiation	RadAvg		889	1160	1526	1846	2197	2482	2392	2224	1873	1414	990	791
	RadStd		62	97	116	138	97	115	79	90	56	58	72	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (5 neighbors)		N/A	N/A	-2.21	N/A	-1.04	-1.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page